

Cues, Partisanship, and Willingness-to-Pay for Energy

Matthew C. Nowlin*

Introduction

Shifting from fossil fuels sources of energy to renewables is needed to mitigate climate change, and such a shift requires the commitment and support of the public. In general, the US public seems to support using more renewable sources, yet there are partisan differences in the degree of that support with Republicans tending to be more supportive of fossil fuels and Democrats tending to be more supportive of renewables (Bergquist, Konisky, and Kotcher 2020; Hamilton et al. 2018; Hawes and Nowlin 2022). Additionally, in the last several years, Republicans have become more supportive of fossil fuels and less supportive of renewables (Kennedy and Kikuchi 2026).

Cues have been shown to drive public opinion on climate change and energy preferences. For example, partisan and elite cues played a significant role in shaping opinion on climate change (Merkley and Stecula 2021; Tesler 2018), and a climate change cue, or frame, can influence energy preferences (Feldman and Hart 2018). In light of recent rhetoric and actions on the part of the second Trump regarding fossil and renewables, it is possible that polarization among the public on energy sources may increase in response. Here, I explore how exposure to a Trump or climate cue, compared to no cue, influences how the US public prioritizes fossil fuels versus renewables as well as their willingness-to-pay (WTP) for those energy sources.

In brief, the literature on WTP for energy finds that the public is responsive to price volatility for both fossil fuels and renewables (Cardella, Ewing, and Williams 2017), but in general are willing to pay more for renewable sources (Borchers, Duke, and Parsons 2007; Nkansah and Collins 2019) and is WTP for research and development by the federal government aimed at reducing the use of fossil fuels (Li et al. 2009). However, concern about the environmental impacts of energy (Knapp et al. 2020) and political beliefs (Aldy, Kotchen, and Leiserowitz 2012) can shape the WTP for energy.

* Associate Professor, Department of Political Science, University of Texas at Arlington. Email: matthew.nowlin@uta.edu

Cues and Public Opinion

Cues shape public opinion on climate and energy issues, often independent of a policy's substantive content. Partisan and elite cues are a particularly strong driver of this process, as citizens frequently adopt the position associated with their party's leaders rather than evaluate a policy on its own terms (Merkley and Stecula 2021; Tesler 2018). Feldman and Hart (2018) found that framing a low-carbon energy policy as a climate change solution, rather than as an air pollution or energy security measure, reduced support for that policy among Republicans specifically, and argued the pattern reflects motivated reasoning rather than a simple heuristic response to the word climate. This distinction matters here because a climate cue and a cue tied to a specific political figure likely activate different cognitive processes, and so may not move opinion in parallel ways.

Partisanship's influence likely extends beyond stated policy support to WTP. Kotchen, Boyle, and Leiserowitz (2013) found that Republicans reported lower WTP for a national climate policy than Democrats and independents, but that this partisan gap was largely explained by differences in belief that climate change is occurring, suggesting that cues capable of shifting climate beliefs could narrow WTP differences across party lines. Using a similar national WTP measure, Kotchen, Turk, and Leiserowitz (2017) estimated an average household WTP of \$177 per year for a carbon tax, with respondents favoring that the revenue be spent on clean energy investment over other uses. More recently, Mayer and Smith (2023) argued that partisanship's effect on climate policy support is conditional on the strength of both in-group identification and out-group threat, and Benegal and Green (2024) found that partisan cue-taking, meaning support that tracks which party proposed a policy, outweighed sensitivity to the policy's stated cost. Scheuch (2026) similarly found partisanship to be among the strongest predictors of WTP for local climate policy, alongside age and income. As expected from this literature, political identity likely conditions not just abstract support for climate and energy policy, but the concrete dollar amounts people report they are willing to pay.

The Public's Willingness-to-Pay for Energy

A separate body of work estimates public WTP for renewable relative to fossil fuel electricity directly, generally without a partisan or climate framing manipulation. Borchers, Duke, and Parsons (2007) found that WTP for green energy differs depending on the renewable source being offered, and Nkansah and Collins (2019) compared WTP for wind against natural gas generation specifically and found consumers willing to pay a premium for wind. Murakami et al. (2015) placed fossil fuels, renewables, and nuclear power on the same WTP scale in the US and Japan, and found that respondents in both countries expressed positive WTP for increased renewable generation but negative WTP for increased nuclear generation, with US respondents expressing a stronger preference for renewables than Japanese respondents did.

A smaller number of studies benchmark WTP for renewables directly against WTP for a fossil fuel alternative, and both likely point toward the identity-conditional pattern examined here. Bakkensen and Schuler (2020) found that Vietnamese households were willing to pay 95 percent more on their monthly electricity bill for reliable renewable power, but were also willing to pay 62 percent more for reliable coal power, and argued that citizens default to supporting fossil fuels unless a renewable alternative is explicitly offered to them. Walter, Thacher, and Chermak (2023) found that US respondents' preferences over fossil versus renewable

energy futures depended on their geographic proximity to fossil fuel production and their opinion about anthropogenic climate change, with the impact of these factors asymmetric across respondent groups. This is among the few sets of studies to compare WTP for multiple energy sources within a single design, an approach extended here to fossil fuels and renewables under experimentally manipulated cues.

Li et al. (2009) estimated household WTP for a federal energy research and development fund aimed at reducing reliance on fossil fuels, and found broad support for that funding even though the policy did not name renewables specifically, suggesting that WTP responds to reducing fossil fuel use as a goal in itself. However, WTP for energy is not solely a function of environmental preference. Cardella, Ewing, and Williams (2017) found that price volatility, not just price level, shapes residential customers' generation-source decisions, and Knapp et al. (2020) found that stated WTP for green electricity programs does not consistently translate into enrollment once a real premium is offered. Aldy, Kotchen, and Leiserowitz (2012) likewise found that WTP for a national clean energy standard depended on respondents' political affiliation as well as on how the standard was structured. Taken together, this literature establishes that the public likely pays a premium for renewable electricity on average, but that the size of that premium, and whether it survives contact with real costs, depends on the source being compared, the framing of the request, and the respondent's political identity, three factors that remain largely unexamined together within a single experimental design.

Methodology and Data

Given the importance of cues and political beliefs in energy preferences, here I use a survey experiment to examine respondents' prioritization of fossil fuels versus renewables as well as their willingness-to-pay (WTP) for those sources to generate electricity in response to a Trump cue or a climate change cue. Specifically, I surveyed 3113 US residents in April 2026, who were randomly assigned to receive one of three treatments, including a cue about Trump's critique's of renewable energy and his support for fossil fuels ($n = 1049$), a climate change cue that stated scientists recommend moving away from fossil fuels to renewables to address climate change ($n = 1013$), or a control condition with no cue ($n = 1051$).

Overall, I expect there to be both general and partisan shifts in preferences based on the cues. In general, those who receive the Trump cue will prioritize fossil and fuels and be willing to pay more, whereas those in the climate condition will prioritize renewable sources and be willing to pay more than those in the other conditions. For political beliefs, I expect conservative Republicans to be responsive to the Trump cue and liberal Democrats to be responsive to the climate cue in their energy preferences.

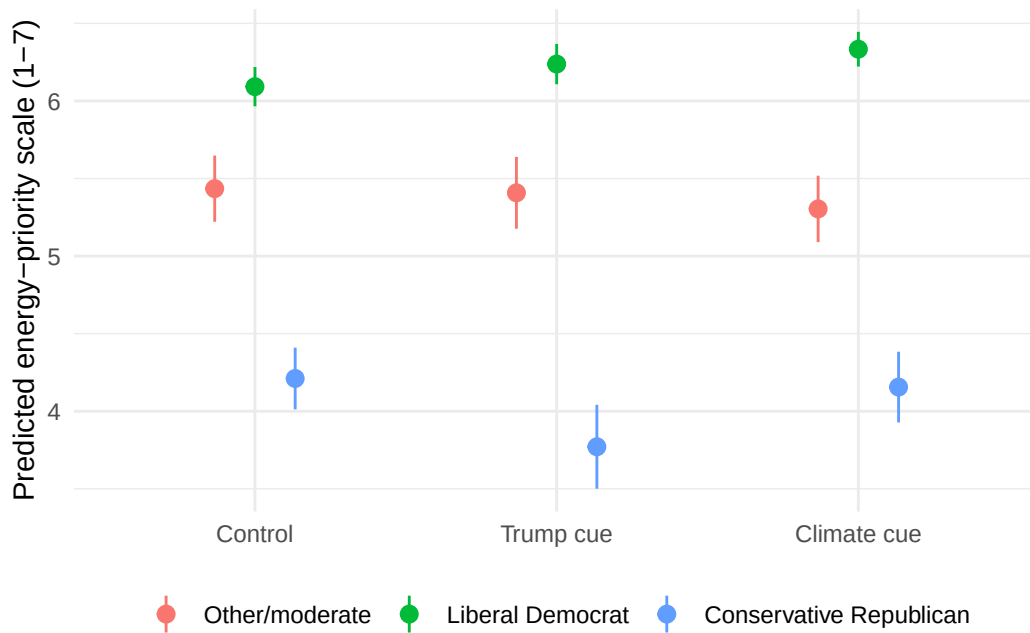
I first examine how respondents chose to prioritize fossil fuels versus renewables. After receiving the cues, respondents were asked, on a 1 to 7 scale, if fossil fuels should be prioritized ($= 1$), fossil fuels and renewables should be prioritized equally ($= 4$), or renewable sources should be prioritized ($= 7$). The mean on the energy-priority scale was 5.17, indicating that respondents, on average, prioritized renewables. Next, I use OLS to examine the prioritization respondents gave to fossil fuels versus renewables as a function of the cue, their political beliefs, and the interaction between cues and political beliefs, controlling for several demographic variables, including age, male, white, education, and income. Energy affordability was a salient concern during the time data was collected: in the survey respondents were asked, on a 0 to 10 scale, their overall concern about the cost of electricity, and the mean was 7.88, indicating a fair amount of concern. Because this concern could

plausibly confound cue and political-identity effects on prioritization and WTP, I re-estimate every model below with concern about energy cost added as an additional control; that specification is reported in the Supplemental Materials. The coefficients, significance, and substantive conclusions are consistent across both specifications.

Results and Discussion

The results are shown in Table 1.

Figure 1: Predicted energy source-priority scale (1-7) by cue condition and political identity (survey-weighted OLS predictions with 95% CIs; age, male, white, education, and income held at their survey-weighted means)



As expected, conservative Republicans were more likely to prioritize fossil fuels than renewables, with an estimated coefficient of -1.22 on the 1 to 7 scale. Liberal Democrats prioritized renewables at 0.66. In terms of the interactions, conservative Republicans who received the Trump cue were more likely to prioritize fossil fuels ($b = -0.41$, $p = 0.077$), whereas liberal Democrats in the climate condition were more likely to prioritize renewables ($b = 0.37$). Overall, the predicted value on the 1 to 7 scale for conservative Republicans in the control group was 4.21, indicating a priority split roughly evenly between fossil fuels and renewables, and it was 3.77 for those receiving the Trump cue, indicating a significant shift toward prioritizing fossil fuels. For liberal Democrats the predicted value was 6.09, indicating a priority for renewables, and the value was 6.33 in the climate condition. The interactions of cue x political beliefs was significant; however, the cues themselves had no effect, on average, on energy source priority compared to the control condition.

Table 1: Effect of cue condition, political beliefs, and their interaction on the energy source-priority scale (survey-weighted OLS; control condition and other/moderate identity as reference)

	Priority scale
Trump cue	−0.027 (0.160)
Climate cue	−0.131 (0.154)
Liberal Democrat	0.657*** (0.126)
Conservative Republican	−1.224*** (0.149)
Trump cue × Liberal Democrat	0.173 (0.184)
Climate cue × Liberal Democrat	0.372** (0.176)
Trump cue × Conservative Republican	−0.413* (0.234)
Climate cue × Conservative Republican	0.075 (0.218)
Age	−0.010*** (0.002)
Male	0.226*** (0.068)
White	0.145* (0.078)
Education	0.041** (0.019)
Income	0.018 (0.013)
(Intercept)	5.493*** (0.162)
Num.Obs.	3113

* p < 0.1, ** p < 0.05, *** p < 0.01

Next, I asked respondents their WTP for fossil fuel and renewable sources using the Gabor-Granger method (e.g., Gabor and Granger 1966). Specifically, I first give respondents information about potential costs associated with each energy source. In the WTP for fossil fuels question, I noted that costs associated with renewable energy have decreased and, therefore they may need to pay more to use fossil fuels. For renewable sources respondents were told about the need to increase distribution. All respondents were asked their WTP for both sources, and the sources were presented in random order. To elicit the maximum/minimum WTP, respondents were shown a random dollar amount between \$1 and \$46, in five-dollar increments, and asked if they would be WTP that amount extra per-month in their electricity bill. If they indicated yes they were shown a larger, random dollar-amount, and if they indicated no they were shown a smaller dollar-amount. This continues until they reach the \$46-maximum or \$1-minimum. Respondents who indicated no at \$1 were coded with a WTP of 0. The percent of respondents WTP each possible dollar amount for each source is shown in Table 2.

Table 2: Distribution of willingness-to-pay for fossil fuel and renewable electricity (weighted percent of respondents choosing each bid amount)

WTP (\$)	Fossil fuels (%)	Renewables (%)
0	47.0	14.0
1	14.7	15.6
6	10.6	14.0
11	5.6	9.1
16	4.9	7.3
21	2.7	4.3
26	2.9	5.8
31	1.2	3.2
36	2.1	4.7
41	1.9	3.8
46	6.3	18.3

Nearly half of respondents, 47 percent, were not WTP any additional amount for fossil fuels, whereas 18.3 percent were WTP the max, \$46 for renewables. Overall, the median WTP for fossil fuels was \$1 with a mean of \$8.33. For renewables the median WTP was \$11 and the mean was \$18.21. Comparing the differences in median and means, the median difference in WTP for renewables versus fossil fuels was \$10. For the mean, the difference was \$9.88, which is significant based on a survey-weighted paired t-test ($t = 22.9$, $df = 3111$, $p < .001$).

Given the large percentage of respondents at \$0 for fossil fuels, I model the WTP for fossil fuels with a two-part hurdle model. First, I use a logit model to see whether a respondent has any positive WTP, then I use OLS to examine differences in WTP amounts. Next, using OLS, I examine the WTP for fossil fuels, among those who are WTP >0, and for renewables. The results of all three models, controlling for demographics, are shown in Table 3.

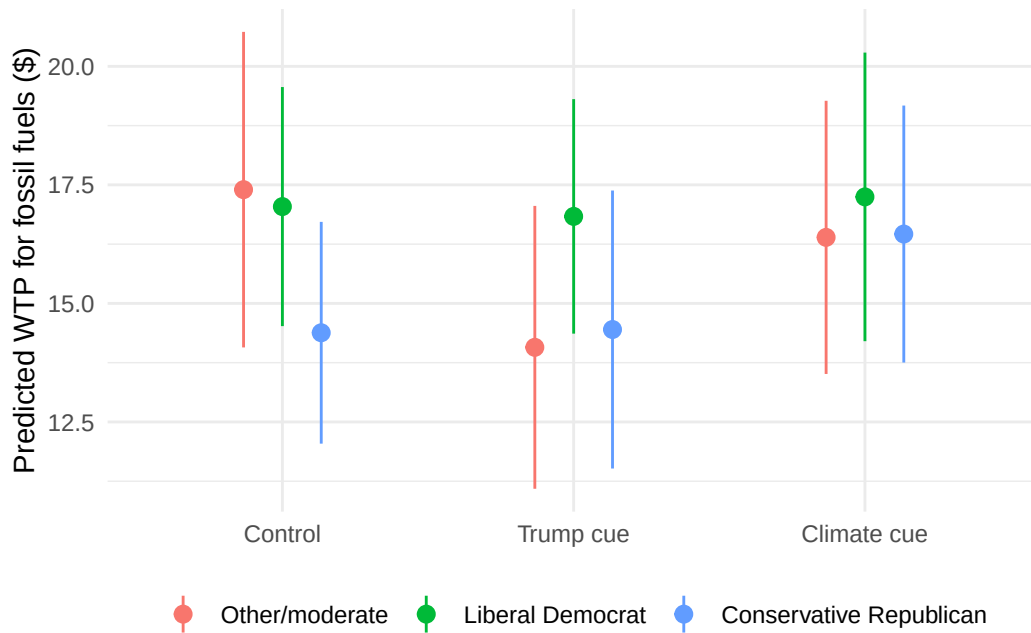
Table 3: Effect of cue condition, political beliefs, and their interaction on willingness-to-pay for fossil fuels (any positive WTP, and amount among those with any positive WTP) and for renewables (survey-weighted logit/OLS; control condition and other/moderate identity as reference)

	Fossil fuels: Any WTP (logit)	Fossil fuels: Amount, if WTP > \$o (OLS)	Renewables
Trump cue	−0.257 (0.210)	−3.326 (2.268)	
Climate cue	0.088 (0.209)	−1.007 (2.231)	
Liberal Democrat	−0.377** (0.186)	−0.357 (2.116)	
Conservative Republican	0.743*** (0.209)	−3.018 (2.091)	
Trump cue × Liberal Democrat	−0.024 (0.261)	3.119 (2.875)	
Climate cue × Liberal Democrat	−0.302 (0.262)	1.211 (2.977)	
Trump cue × Conservative Republican	0.425 (0.302)	3.394 (2.937)	
Climate cue × Conservative Republican	−0.044 (0.296)	3.088 (2.886)	
Age	0.007** (0.003)	−0.015 (0.033)	
Male	−0.042 (0.093)	−0.180 (1.026)	
White	−0.261** (0.106)	0.959 (1.201)	
Education	−0.023 (0.027)	1.166*** (0.291)	
Income	−0.031* (0.019)	0.310 (0.211)	
(Intercept)	0.104 (0.221)	12.291*** (2.457)	
Num.Obs.	3113	1433	

* p < 0.1, ** p < 0.05, *** p < 0.01

For fossil fuels, conservative Republicans were significantly more likely than the reference group to have any positive WTP ($b = 0.74$, log-odds scale), predicted at 68.4% in the control condition compared to 41.4% for liberal Democrats ($b = -0.38$, $p = 0.043$). However, the cues were not significant nor were the interactions between the cues and political beliefs.

Figure 2: Predicted willingness-to-pay for fossil fuels, among respondents with any positive WTP, by cue condition and political identity (survey-weighted OLS predictions with 95% CIs; age, male, white, education, and income held at their survey-weighted means)

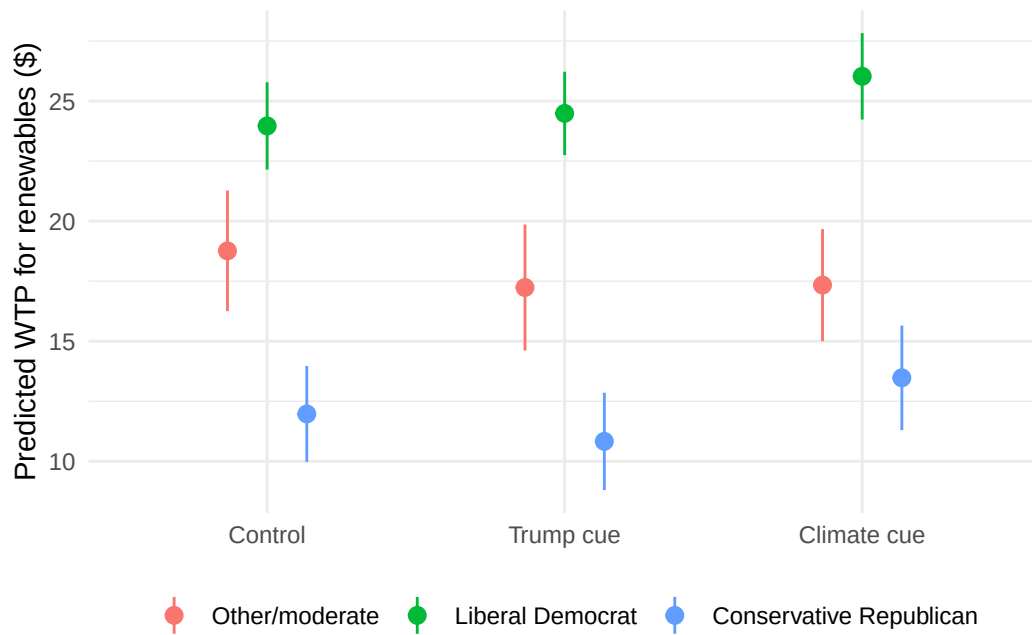


Neither the Trump cue nor its interaction with conservative Republican identity significantly predicted participation or, among those with positive WTP, the dollar amount: predicted participation for conservative Republicans was 68.4% in the control condition versus 71.9% in the Trump cue, and the predicted amount conditional on paying something was \$14.38 versus \$14.45.

Looking to renewables, liberal Democrats were willing to pay \$5.20 more, on average, and in the climate condition the amount increased further by \$3.49 ($p = 0.109$), for a total predicted willingness to pay of \$26.03 for renewables. Conservative Republicans, however, were willing to pay \$6.79 less for renewables, for a predicted total of \$11.97.

The clean energy transition is needed to mitigate climate change, and public support is needed for the energy transition to complete. The results here show that support for fossil fuel and renewable energy sources is polarized, as measured both by prioritizing one source over the other as well as WTP for those sources. Additionally, a Trump cue made conservative Republicans more supportive of fossil fuels and a climate cue made liberal Democrats more supportive of renewables. However, respondents were, on average, more support

Figure 3: Predicted willingness-to-pay for renewables by cue condition and political identity (survey-weighted OLS predictions with 95% CIs; age, male, white, education, and income held at their survey-weighted means)



of renewables in terms of both priority and WTP. Even conservative Republicans in the Trump cue had a WTP higher for renewables, \$11.97, than fossil fuels, at \$10.39.

Conclusion and Policy Implications

References

- Aldy, Joseph E., Matthew J. Kotchen, and Anthony A. Leiserowitz. 2012. "Willingness to Pay and Political Support for a US National Clean Energy Standard." *Nature Climate Change* 2(8): 596–99.
- Bakkensen, Laura, and Paul Schuler. 2020. "A Preference for Power: Willingness to Pay for Energy Reliability Versus Fuel Type in Vietnam." *Energy Policy* 144: 111696.
- Benegal, Salil, and Jon Green. 2024. "Cost Sensitivity, Partisan Cues, and Support for the Green New Deal." *Journal of Environmental Studies and Sciences* 14(4): 763–75.
- Bergquist, Parrish, David M Konisky, and John Kotcher. 2020. "Energy Policy and Public Opinion: Patterns, Trends and Future Directions." *Progress in Energy* 2(3): 032003.
- Borchers, Allison M., Joshua M. Duke, and George R. Parsons. 2007. "Does Willingness to Pay for Green Energy Differ by Source?" *Energy Policy* 35(6): 3327–34.
- Cardella, Eric, Bradley T. Ewing, and Ryan B. Williams. 2017. "Price Volatility and Residential Electricity Decisions: Experimental Evidence on the Convergence of Energy Generating Source." *Energy Economics* 62: 428–37.
- Feldman, Lauren, and P. Sol Hart. 2018. "Climate Change as a Polarizing Cue: Framing Effects on Public Support for Low-Carbon Energy Policies." *Global Environmental Change* 51: 54–66.
- Gabor, André, and C. W. J. Granger. 1966. "Price as an Indicator of Quality: Report on an Enquiry." *Economica* 33: 43–70.
- Hamilton, Lawrence C., Erin Bell, Joel Hartter, and Jonathan D. Salerno. 2018. "A Change in the Wind? US Public Views on Renewable Energy and Climate Compared." *Energy, Sustainability and Society* 8(1): 11.
- Hawes, Rachel, and Matthew C. Nowlin. 2022. "Climate Science or Politics? Disentangling the Roles of Citizen Beliefs and Support for Energy in the United States." *Energy Research & Social Science* 85: 102419.
- Kennedy, Brian, and Emma Kikuchi. 2026. "Americans' Shifting Views on Energy Sources, Policy in 2026." <https://www.pewresearch.org/science/2026/04/03/americans-shifting-views-on-energy-issues/> (April 21, 2026).
- Knapp, Lauren, Eric O'Shaughnessy, Jenny Heeter, Sarah Mills, and John M. DeCicco. 2020. "Will Consumers Really Pay for Green Electricity? Comparing Stated and Revealed Preferences for Residential Programs in the United States." *Energy Research & Social Science* 65: 101457.
- Kotchen, Matthew J., Kevin J. Boyle, and Anthony A. Leiserowitz. 2013. "Willingness-to-Pay and Policy-

- Instrument Choice for Climate-Change Policy in the United States.” *Energy Policy* 55: 617–25.
- Kotchen, Matthew J., Zachary M. Turk, and Anthony A. Leiserowitz. 2017. “Public Willingness to Pay for a US Carbon Tax and Preferences for Spending the Revenue.” *Environmental Research Letters* 12(9).
- Li, Hui, Hank C. Jenkins-Smith, Carol L. Silva, Robert P. Berrens, and Kerry G. Herron. 2009. “Public Support for Reducing US Reliance on Fossil Fuels: Investigating Household Willingness-to-Pay for Energy Research and Development.” *Ecological Economics* 68(3): 731–42.
- Mayer, Adam P., and E. Keith Smith. 2023. “Multidimensional Partisanship Shapes Climate Policy Support and Behaviours.” *Nature Climate Change* 13(1): 32–39.
- Merkley, Eric, and Dominik A. Stecula. 2021. “Party Cues in the News: Democratic Elites, Republican Backlash, and the Dynamics of Climate Skepticism.” *British Journal of Political Science* 51(4): 1439–56.
- Murakami, Kayo, Takanori Ida, Makoto Tanaka, and Lee Friedman. 2015. “Consumers’ Willingness to Pay for Renewable and Nuclear Energy: A Comparative Analysis Between the US and Japan.” *Energy Economics* 50: 178–89.
- Nkansah, Kofi, and Alan R. Collins. 2019. “Willingness to Pay for Wind Versus Natural Gas Generation of Electricity.” *Agricultural and Resource Economics Review* 48(1): 44–70.
- Scheuch, Eric. 2026. “Voters Support and Are Willing to Pay for Local Climate Adaptation Policies.” *Urban Affairs Review*: 10780874261475229.
- Tesler, Michael. 2018. “Elite Domination of Public Doubts About Climate Change (Not Evolution).” *Political Communication* 35(2): 306–26.
- Walter, Kara A., Jennifer Thacher, and Janie M. Chermak. 2023. “Examining Willingness to Pay for Energy Futures in a Fossil and Renewable Energy-Rich Locale.” *Energy Policy* 181: 113661.